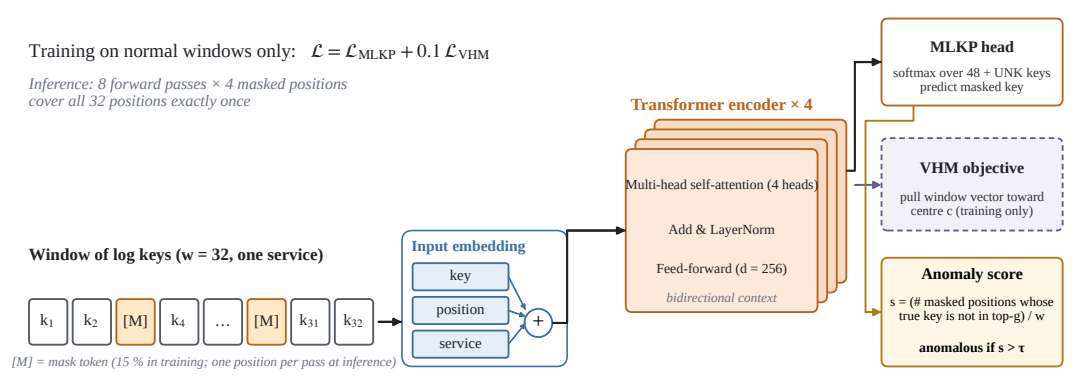
Training on normal windows only: $\mathcal{L} = \mathcal{L}_{\text{MLKP}} + 0.1\,\mathcal{L}_{\text{VHM}}$
Inference: 8 forward passes × 4 masked positions
cover all 32 positions exactly once
Window of log keys (w = 32, one service)
k1
k2
[M]
k4
…
[M]
k31
k32
[M] = mask token (15 % in training; one position per pass at inference)
Input embedding
key
position
service
+
Transformer encoder × 4
Multi-head self-attention (4 heads)
Add & LayerNorm
Feed-forward (d = 256)
bidirectional context
MLKP head
softmax over 48 + UNK keys
predict masked key
VHM objective
pull window vector toward
centre c (training only)
Anomaly score
s = (# masked positions whose
true key is not in top-g) / w
anomalous if s > τ